30 December 1968
SUPERSEDING
PA-PD-2532
15 February 1965

MILITARY SPECIFICATION

WRENCH, SPANNER: H-4194A

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of spanner wrench used in assembling fuze to XM-454 and XM-455 projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Fiberboard

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MS-16535-130 - Rivet

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8857571 - Wrench, Spanner H-4194A
8864056 - Carton, Packing, Ammunition for Wrench, Spanner H-4194A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

8864707 - Grip
8864710 - Wrench, Spanner
ET-8857571 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly. The wrench shall comply with all requirements specified on Drawing (dwg.) 8857571, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and Metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Major characteristic table I		Minor characteristic table II	
Lot size	Sample size	Lot size	Sample size
1- 155	100% inspection	1- 25	100% inspection
156- 200	155	26- 30	25
201- 250	175	31- 35	26
251- 300	190	36- 40	27
301- 350	200	41- 45	28
351- 400	210	46- 50	29
401- 450	220	51- 55	30
451- 500	225	56- 60	31
501- 550	230	61- 65	32
551- 600	235	66- 70	33
601- 650	240	71- 75	34
651- 700	243	76- 85	35
701- 750	247	86- 95	36
751- 800	250	96- 110	37
801- 900	254	111- 130	38
901-1000	258	131- 156	39
1001-1100	261	157- 190	40
1101-1200	264	191- 240	41
1201-1300	266	241- 340	42
1301-1400	268	341- 520	43
1401-1500	270	521-1120	44
1501-1600	272	1121 and over	45

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

4.3.2.1.1 Wrench, Spanner (see dwg 8864710 covering a detail of dwg 8857571).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Width of span	Gage
102.	Straight length inside of span, min	Gage
103.	Thickness of spanner	Gage
104.	Flatness of wrench	Gage
105.	Rockwell hardness	Gage
106.	Materials (see 3.1)	Visual
Minor:		
201.	Total width of spanner, min	Gage
202.	Width of outside ends of spanner, min	Gage
203.	Depth of span, min	Gage
204.	Width of handle, min	Gage
205.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8864710	Visual
206.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.2 Grip (see dwg 8864707 covering a detail of dwg 8857571).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Length, min	Gage
202.	Width, min	Gage
203.	Thickness, min	Gage
204.	Split, crack of gouge	Visual
205.	Objective evidence (see 6.2) available for protective finish in accordance with dwg 8857571	Visual
206.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.3 Wrench, Spanner H4194A (see dwg 8857571).

Categories	Characteristics	Method of inspection
Critical: None defined.		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

Major:
101. Objective evidence (see 6.2) available for rivet in accordance with MS-16535-130.....Visual
102. No rivets loose, missing, sharp edges or cracked.....Visual/Manual
103. No grips cracked, split, rough gripping surfaces, and overlap of metal handle of wrench.....Visual/Manual

Minor:
201. Stamping.....Visual
202. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.4 Carton, Packing, Ammunition, for Wrench, spanner H-4194A (see dwg 8864056 and Section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Wrench tight in carbon.....Manual
102. Packaging materials (see 3.1).....Visual
103. Sealing in accordance with PPP-B-636.....Visual
104. Label waterproofed.....Visual

Minor:
201. Marking.....Visual
202. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.5 Packing and Marking for Storage and Shipments (see dwg 8808362 and Section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Gross load weight, max.....*Measure
102. Contents tight in package.....Manual
103. Correctly packed.....Visual/Manual
104. Package material (see 3.1).....Visual

* Determine weight with a suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.3).....Visual/Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

4.3.3 Testing. Not applicable.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8857571. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. Not applicable.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The wrench shall be packaged in accordance with dwg 8864056 and 8808362 using the specified moisture barrier.

5.1.2 Level C. The wrench shall be packaged in accordance with dwg 8864056 and 8808362 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The wrench shall be packed in accordance with dwg 8864056 and 8808362.

5.3 Marking. Marking shall be as specified on dwgs 8864052 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
Army - MU
Project No. 1190-A010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-W-50222(MU)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6279

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:17Z
Check the source to verify that this is the current version before use.